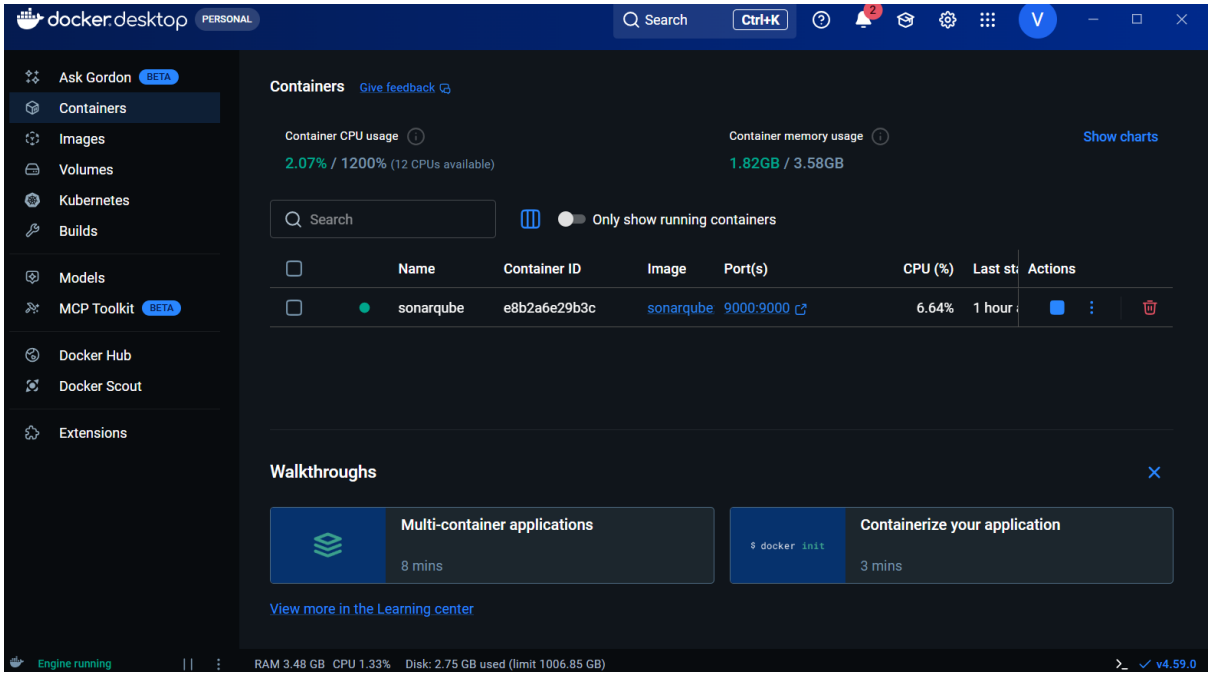
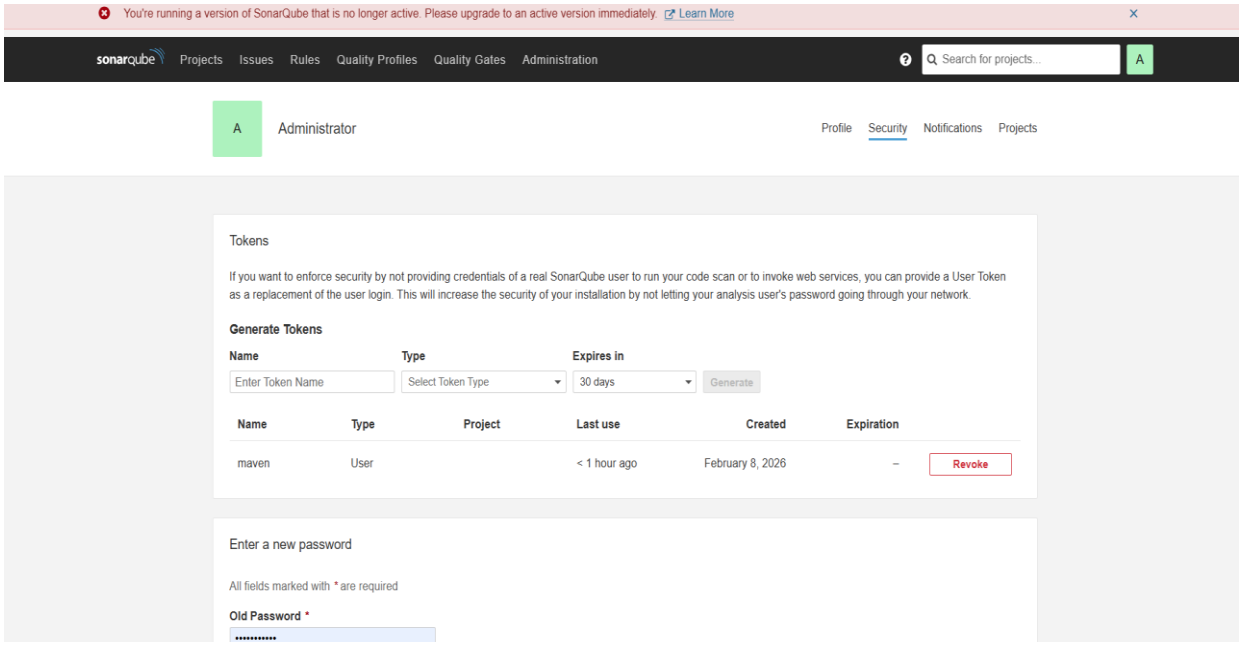
**Prerequisites**:

1. Java intalled version 17+
2. Apache maven 4.0(used)
3. Docker desktop
4. **docker run -d --name sonarqube -p 9000:9000 sonarqube:lts**
   use this command to run SonarQube as docker container




5. As port is set to 9000 access http://localhost:9000 and setup credentials
6. Generate Token by choosing your token name and type with required expiration type in security tab




7. Create Maven project is any of the IDE

8. Configure pom.xml and add properties (YOUR_TOKEN = generated token on **localhost/9000**)

```
<properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>

    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>

    <!-- SonarQube Configuration -->
    <sonar.host.url>http://localhost:9000</sonar.host.url>
    <sonar.java.coveragePlugin>jacoco</sonar.java.coveragePlugin>
    <sonar.coverage.jacoco.xmlReportPaths>
        target/site/jacoco/jacoco.xml
    </sonar.coverage.jacoco.xmlReportPaths>
    <sonar.login>YOUR_TOKEN</sonar.login>
</properties>
```

9. And also add build plugins required for the sonar

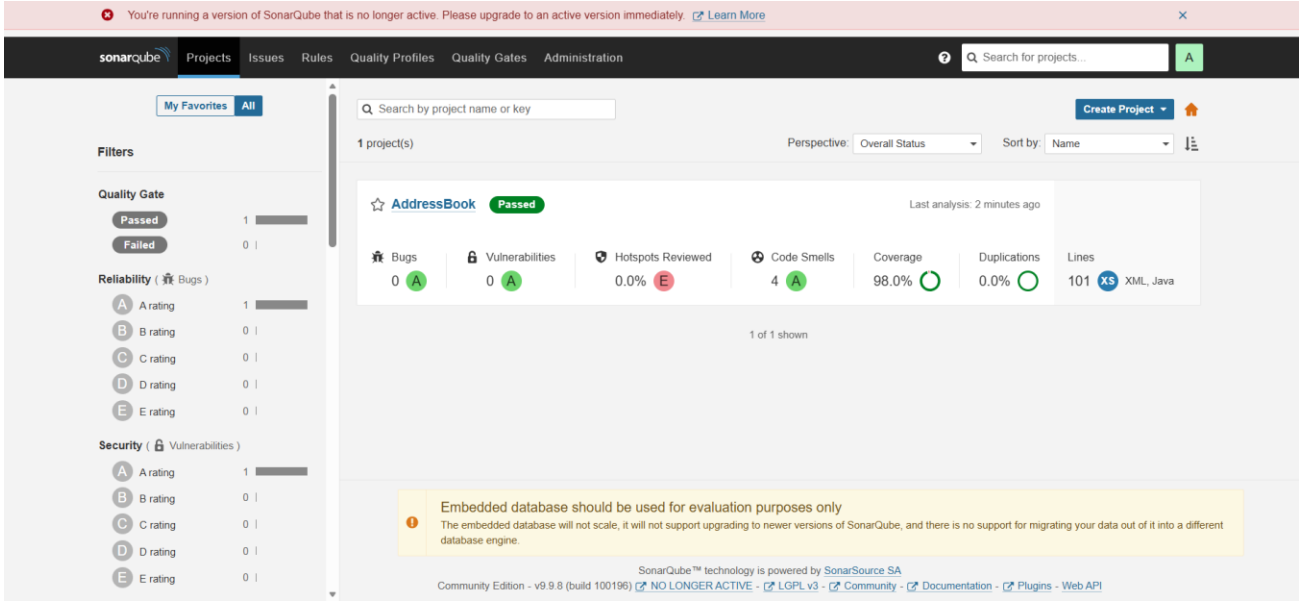
```
<project xmlns="http://maven.apache.org/POM/4.0.0"
    <build>
        <plugins>
            <plugin>
                <executions>

                    <execution>
                        <id>prepare-agent</id>
                        <goals>
                            <goal>prepare-agent</goal>
                        </goals>
                    </execution>

                    <execution>
                        <id>report</id>
                        <phase>test</phase>
                        <goals>
                            <goal>report</goal>
                        </goals>
                    </execution>
                </executions>
            </plugin>
            <plugin>
                <groupId>org.sonarsource.scanner.maven</groupId>
                <artifactId>sonar-maven-plugin</artifactId>
                <version>3.10.0.2594</version>
            </plugin>

        </plugins>
```

**10.** Now run the command of maven to test the code quality using SonarQube
**mvn clear verify sonar:sonar**

**LifeCycle:**
Deletes old build → builds project → runs tests → sends report to SonarQube.

11. Project report on SonarQube dashboard